

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping enterprise AI integrations from healthcare RAG and agentic workflows to financial-data pipelines - ~35% retrieval precision gains, >30% hallucination reduction, ~30% post-deployment defect reduction, and ~30% deployment-error reduction on compliance-sensitive ERP integrations. Builds end-to-end with Python, SQL/T-SQL, FastAPI, Docker, Jenkins CI/CD, and structured logging. Comfortable owning customer-facing implementation work in regulated environments with RBAC, audit logging, and stakeholder translation across technical and business audiences.

CORE COMPETENCIES

AI Forward Deployed delivery • SQL & ETL/ELT pipelines • RAG & agentic workflows • Python & REST APIs • Docker & containerization • Enterprise integrations • RBAC, audit logging & compliance • Customer-facing stakeholder translation

WORK EXPERIENCE

Progress Solutions Inc. - Jr. AI/ML Engineer Trainee

Jul 2025 - Present

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- **Applied AI delivery for enterprise healthcare workflows:** RAG pipelines with recursive semantic chunking and grounded eval harnesses raised contextual retrieval precision ~35%, reduced irrelevant retrieval >30%, with response-alignment above 90%.
- **Agentic LLM workflows** for multi-step queries (eligibility checks, care workflow navigation, documentation clarification) with structured outputs, grounding rules, and instrumented evaluation; lifted agent response stability ~25%.
- **Predictive ML pipelines** (scikit-learn, XGBoost) for no-show prediction, engagement scoring, and risk prioritization; raised recall 15-20% on high-risk categories while holding precision >90%.
- **FastAPI/Flask REST services** containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects ~30%.
- **HIPAA-conscious data governance:** de-identification, lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs - translating model behavior to non-technical clinical leads.

Energy Solutions International (Emerson) - Database & DevOps Performance Engineer (Intern)

Jun 2022 - Apr 2023

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); reduced query execution time ~20% and data retrieval latency ~25%.
- **Jenkins CI/CD pipelines** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors >30% and improved release cycle ~35-40% via structured release validation and rollback checkpoints.
- Performance dashboards using **SQL Server DMVs and Grafana** to track long-running queries, deadlocks, and CPU/I/O contention; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing) and reconciliation queries; improved reconciliation runtime ~15% and reduced deadlocks ~15%.
- Supported **RBAC implementation and audit logging** for financial modules in a compliance-sensitive oil & gas environment.

Suvidha Foundation - Machine Learning Engineer

Jan 2022 - Mar 2022

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; transcript preprocessing, hierarchical summarization, and timestamp-aligned clip extraction reduced manual review time **60-70%** (hours → <5 minutes per video).
- Achieved ~85% **alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** preventing cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Patient Records Management Web Application

Bottle + SQLite relational healthcare CRUD app with normalized **patients/diagnoses** schema, foreign-key constraints, application-layer cascade deletes, and **LEFT JOIN + GROUP_CONCAT** aggregation for single-query patient + diagnosis retrieval.

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams, generates poetic reinterpretations, and synthesizes dreamscapes via Perplexity Sonar (interpretation) + Replicate (16:9 visuals). **Intermediate structured prompt transformation layers** lifted contextual alignment ~**30%** and first-pass image success rate ~**25-30%** over naive direct-prompt orchestration.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (Shipped, Chrome Web Store)

Manifest V3 extension built solo - content scripts, service workers, **captureVisibleTab** for panel capture. **Four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind one abstraction. Multi-section pipeline for tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selectors for 29 manga/webtoon sites, privacy policy, and narrowed host permissions.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - Authorized to work in the United States; future sponsorship required.

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers & Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems